Python Activity-Based Learning
27. Number Pattern Generator

Simple student-style assignment | Easy English | Step-by-step solution

1. Problem Statement

Write a Python program to print a simple right-angled number pattern. The user enters the number of rows, and each row displays numbers from 1 up to the row number.

2. Objectives

• Take the number of rows from the user.

• Use nested loops to print a pattern.

• Practise number output and loop control.

3. Assumptions

• The number of rows must be between 1 and 10.

• The pattern starts from 1 in every row.

4. Algorithm (Step by Step)

1. Start.

2. Read the number of rows.

3. If rows is not between 1 and 10, display Invalid input.

4. Otherwise, use an outer loop for each row.

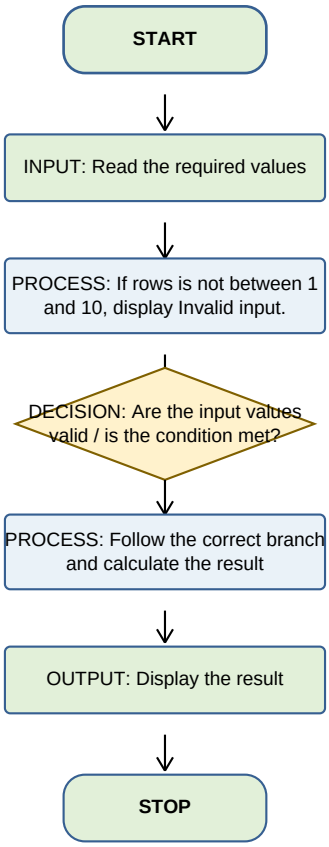
5. Use an inner loop to print numbers from 1 to the current row.

6. Move to the next line after each row.

7. Stop.

5. Flowchart

Follow the arrows from START to STOP. The diamond shows a decision point.



6. Pseudocode

```
START
INPUT rows
IF rows < 1 OR rows > 10 THEN PRINT Invalid input
ELSE
    FOR i = 1 TO rows
        FOR j = 1 TO i
            PRINT j without newline
        PRINT newline
STOP
```

7. Python Source Code

```
rows = int(input("Enter number of rows (1 to 10): "))

if rows < 1 or rows > 10:
    print("Invalid number of rows")
else:
    for i in range(1, rows + 1):
        for j in range(1, i + 1):
            print(j, end=" ")
        print()
```

8. Test Cases

Input / Test	Expected result	What it checks
3	1 / 1 2 / 1 2 3	Three rows
1	1	One row
0	Invalid number of rows	Zero is not allowed

9. Sample Output

```
Enter number of rows (1 to 10): 4
1
1 2
1 2 3
```

```
1 2 3 4
```

10. Conclusion

This program uses nested loops to generate a number pattern based on the number of rows entered by the user.